SINAG 2U CUBESATELLITE STANDARD DESIGN REQUIREMENT FOR THE DEVELOPMENT OF CAD AND EDA

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Can be used for other projects to practice design standards.

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Table 1. Change Log

Revision ID	Document Date	Change Authority	Affected Paragraphs	Comments
001 v.A	Dec 2025	F. Camogao	All	Original

1 INTRODUCTION

This document establishes the foundational framework and technical directives for all **Computer-Aided Design (CAD)** and **Electronic Design Automation (EDA)** activities. The primary objectives are **Compatibility** and **Uniformity**. Adherence to these standards ensures the development of maintainable, high-fidelity models. By eliminating spatial interference and alignment errors, we mitigate the risk of development delays and increased costs. This document serves as the authoritative reference for all design phases, ensuring a seamless transition from digital modeling to physical deployment.

To maintain technical consistency, the following software suites are mandated for the development of electronics, physical structures, and mission-critical assets:

- **Mechanical Design & Drafting:** Autodesk Fusion and AutoCAD. These tools are selected for their robust 3D modeling capabilities, suitability for additive manufacturing (3D printing), and complex assembly management.

- **Electronics Design (EDA):** EasyEDA (Standard and Professional).
  - *Note:* While both versions share a common ecosystem, **EasyEDA Professional** is the preferred standard for final 3D PCB exports due to its superior geometric precision and advanced feature set.

- **Simulation & Analysis:** Ansys. This will be utilized for structural and thermal verification to ensure flight readiness.

  This standard is developed **solely for this project**.

CORE PHILOSOPHY

Our engineering approach is governed by the constraints of a **Safety-Critical** environment and **Time-Sensitive** delivery schedules. Every design choice must reflect the following principles:

- **Defensive Design and Constraints:** Operate under the assumption that components and modules will encounter interference. All tolerances, hole alignments, and mounting interfaces must be explicitly verified and validated within the digital environment before fabrication.

- **Determinism and Complexity Management:** Prioritize predictable fabrication timelines. Designers must maximize internal volume efficiency and simplify installation procedures to reduce the "mean time to repair" or assembly.

- **Visual Uniformity and Execution:** All structural housings, PCB layouts, and mechanical interfaces must exhibit a professional, uniform aesthetic. Design for "Deployment Readiness" means the asset must be functional, robust, and aesthetically compliant with aerospace industry standards.

2 REFERENCED DOCUMENTS

N/A

3 GENERAL DESIGN

The purpose of this standard document is intended to help the onboard computer developers and other team members who might or may involved in the process of coding to achieve their respective team objectives. This helps the programmers to develop code that conforms to safety-critical software principles, i.e., code that does not contain defects that could lead to catastrophic failures resulting in significant harm to individuals and/or equipment. In general, the code produced should exhibit the following important qualities:

Reliability: Executable code should consistently fulfill all requirements in a predictable manner.

Portability: Source code should be portable (i.e. not compiler or linker dependent).

Maintainability: Source code should be written in a manner that is consistent, readable, simple in design, and easy to debug.

Testability: Source code should be written to facilitate testability. Minimizing the following characteristics for each software module will facilitate a more testable and maintainable module:

1. **code size**
2. **Complexity**
3. **static path count** (number of paths through a piece of code)

Reusability: The design of reusable components is encouraged. Component reuse can eliminate redundant development and test activities (i.e. reduce costs).

Extensibility: Requirements are expected to evolve over the life of a product. Thus, a system should be developed in an extensible manner (i.e. perturbations in requirements may be managed through local extensions rather than wholesale modifications).

Readability: Source code should be written in a manner that is easy to read, understand and comprehend.

Note that following the guidelines contained within this document will not guarantee the production of an error-free, safe product. However, adherence to these guidelines, will help programmers produce clean designs that minimize common sources of mistakes and errors.

** Some following texts are directly copied from the JSF-AV Rules PDF but some are revised/modified to fit to the context of this project.*

4 EDA STANDARDS (STRICT)

Severity: MANDATORY. Design that is not aligned will be rejected during the review process.

4.1 Schematic
4.2 Pin Configurations
4.3 Protection and Regulation
4.4 Filters
4.5 Output
4.6 Calculations
4.7 PCB Placement and Nets
4.8 PCB Layout and Thickness
4.9 PCB DRC and ERC
4.10 PCB Export

5 CAD STANDARDS (STRICT)

5.1 Dimensions and Constrains
5.2 Holes and Screws to Use
5.3 Mechanisms and Limitations
5.4 Locations and Placement
5.5 Placement and Installation Procedures
5.6 Component Housing and Protection

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8 CHECKLISTS

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